

1 *nf-genetic-correlations*: a Nextflow pipeline to perform 2 global and local genetic correlations

3 **Frida Lona-Durazo** ¹, **Fatima Barry**^{1,2}, and **Sarah A. Gagliano Taliun**^{1,3,4}

4 **1** Institut de Cardiologie de Montréal, Montreal, Quebec, Canada **2** Université de Montréal, Quebec,
5 Canada **3** Department of Medicine, Faculty of Medicine, Université de Montréal, Quebec, Canada **4**
6 Department of Neurosciences, Faculty of Medicine, Université de Montréal, Quebec, Canada

DOI: [10.xxxxxx/draft](https://doi.org/10.xxxxxx/draft)

Software

- [Review](#) 
- [Repository](#) 
- [Archive](#) 

Editor: [Open Journals](#) 

Reviewers:

- [@openjournals](#)

Submitted: 01 January 1970

Published: unpublished

License

Authors of papers retain copyright
and release the work under a
Creative Commons Attribution 4.0
International License ([CC BY 4.0](#)).

7 Summary

8 Complex human traits and diseases are in part influenced by genetic variants across the full
9 DNA sequence (the genome). Genome-wide association studies (GWAS) identify which genetic
10 regions (loci) are statistically associated with the trait of interest. Many genomic regions or
11 genes are associated with multiple traits, which is referred to as pleiotropy. Assessing genetic
12 correlations between two traits (i.e. pairwise correlation at the level of genetic associations), can
13 help identify pleiotropic regions, and provide etiological insights into shared biology between
14 the tested traits ([Bulik-Sullivan et al., 2015](#); [Van Rheenen et al., 2019](#)). Correlations can
15 be assessed genome-wide (global genetic correlation) or at a locus by locus basis (local
16 genetic correlation). The widespread availability of GWAS summary-level results has resulted
17 in the possibility of testing the global and local genetic correlations across thousands of
18 polygenic traits to uncover pleiotropy, using bioinformatic tools that require several inputs and
19 varying formats of GWAS summary statistics. To facilitate the estimation of both global and
20 local genetic correlations using two of the most common bioinformatic tools (i.e., Linkage
21 Disequilibrium Score Regression [LDSC] and Local Analysis of covariant Association [LAVA])
22 ([Bulik-Sullivan et al., 2015](#); [Werme et al., 2022](#)), we developed a Nextflow workflow named
23 *nf-genetic-correlations*, to automate both approaches. Additionally, our workflow also allows
24 to perform genetic colocalization as a follow-up analysis, which aids to uncover shared causal
25 variants within a genetic region across pairs of traits.

26 Statement of need

27 The *nf-genetic-correlations* Nextflow pipeline performs global genetic correlations with LDSC
28 and local genetic correlations with LAVA across as many traits as the user requires (minimum
29 two traits). These two bioinformatic tools are widely and systematically used in human genetics
30 research, but, to our knowledge, there is not an easy-to-use approach that combines both into
31 one streamlined pipeline ([Bright et al., 2026](#); [Hu et al., 2024](#); [Kim et al., 2024](#); [Liwayiding et al., 2025](#); [Lona-Durazo et al., 2023](#); [Reynolds et al., 2023](#); [Wu et al., 2026](#); [Zhang et al., 2025](#)).
32 LDSC was originally built under Python 2, whereas LAVA is an R package. One of the main
33 advantages of *nf-genetic-correlations* is that both LDSC and LAVA are downloaded by users
34 as Apptainer containers with all dependencies needed, without the need of installing external
35 libraries or dealing with version compatibility. Additionally, LAVA is automatically parallelized
36 within our pipeline, which significantly decreases the usual running time, without the users
37 needing to adapt their R scripts for the same purpose. In terms of the inputs, the user only
38 needs to provide the GWAS summary statistics in a standard format (i.e., we use the column
39 names as in the GWAS Catalog) and a metadata file that is used throughout the pipeline.
40 Therefore, *nf-genetic-correlations* facilitates the work when performing genetic correlations
41 across multiple traits, and outputs the customary readable text files for both LDSC and LAVA.
42

43 Additionally, the pipeline can optionally streamline genetic colocalizations with the R package
 44 *coloc* (Giambartolomei et al., 2014), which fully eliminates the installation, as well as the data
 45 and custom scripts preparation steps. Overall, *nf-genetic-correlations* reduces approximately
 46 ten times the computational time of the LAVA process by utilizing parallelization. Additionally,
 47 the pipeline is user-friendly, versatile, well documented and can be executed across different
 48 systems and environments, all of which enables users with different levels of bioinformatics
 49 expertise to perform global and local genetic correlations.

50 State of the field

51 To our knowledge, the field lacks a unified framework that streamlines global and local genetic
 52 correlations, along with genetic colocalization into one automated pipeline, using two of
 53 the most common bioinformatics tools available for genetic correlations: LDSC and LAVA.
 54 Our nextflow pipeline *nf-genetic-correlations* aims to fill this gap. Previous published studies
 55 (including our own work) (Bright et al., 2026; Hu et al., 2024; Kim et al., 2024; Liwayiding
 56 et al., 2025; Lona-Durazo et al., 2023; Reynolds et al., 2023; Wu et al., 2026; Zhang et al.,
 57 2025) that have implemented both LDSC and LAVA used both approaches separately, which
 58 works well for a one-time specific project. However, in a research environment where GWAS
 59 results are constantly being updated and analyses need to be run many times across several
 60 traits, as is common in various human and population genetics research groups, a pipeline
 61 like *nf-genetic-correlations* becomes a practical option to quickly obtain results and reduce
 62 scripting and debugging times.

63 Software design

64 *nf-genetic-correlations* was developed using Nextflow, one of the most widely used workflow
 65 management tools in bioinformatics, which allows the use of multiple programming languages
 66 interchangeably, including Python and R. Each process in the workflow is automatically run
 67 after the successful completion of the previous one, without requiring the user to intervene in
 68 between steps. A summary of the workflow is presented in Figure 1, and it consists on the
 69 following architecture:

- 70 1) Formatting the GWAS summary statistics based on the metadata file that the user inputs,
 71 to be used for LDSC, LAVA and optionally colocalization.
- 72 2) Munging the GWAS summary statistics for LDSC using the LDSC custom Python 2
 73 script.
- 74 3) Estimating the genetic heritability of each of the input traits with LDSC, followed by
 75 estimating the pairwise global genetic correlation with LDSC across all possible traits
 76 pairs.
- 77 4) Preparing the sample overlap and metadata files for LAVA, which is generated after LDSC
 78 is completed (i.e. cannot be run in parallel with LDSC). LAVA requires the covariance
 79 estimates derived from LDSC to account for sample overlap in its assessment of local
 80 genetic correlations.
- 81 5) Estimating local genetic correlations with LAVA (univariate and bivariate tests) in a
 82 parallelized manner across all pairs of traits, where 2,495 genomic regions are partitioned
 83 in 10 equal parts, as they are independent from each other. Importantly, the LAVA
 84 bivariate test is performed only for those pairs of traits which univariate test at that
 85 genomic region was significant using a Bonferroni-correction ($p\text{-value} < 0.05/2,945$)
 86 as it is suggested in the original LAVA publication (Werme et al., 2022). Additionally,
 87 the linkage disequilibrium (LD) reference used for LAVA is by default the UK Biobank
 88 reference (as recently suggested by LAVA developers) (De Leeuw et al., 2025), but the

89 user can change it to use the 1000 Genomes EUR LD reference. When all ten LAVA
90 partitions are completed, they will be merged into a single output readable file.

91 6) If the user opted to perform genetic colocalizations, the process begins for those pairs
92 of traits which were significant in the LAVA bivariate test. By default, the significance
93 threshold is defined by having a p -value < 0.05 , but the user can adjust the threshold.

94 7) Finally, intermediate files are deleted and only the final results from each of the main
95 steps of the pipeline are retained: LDSC heritability and global genetic correlations, LAVA
96 univariate and bivariate tests, and, optionally, genetic colocalization results. However,
97 the user can opt to keep intermediate files as well.

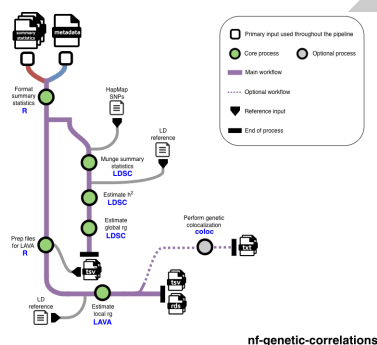


Figure 1: Figure 1. Workflow of *nf-genetic-correlations* processes

98 The pipeline is designed so that users do not need to edit or adapt the Nextflow file (*main_full.nf*)
99 where the workflow and all processes are described. Instead, the user can change certain
100 settings by using optional flags in the executed Nextflow command. Therefore, users do not
101 need to be proficient in Nextflow to use *nf-genetic-correlations*. The Nextflow configuration
102 file (*nextflow.config*) includes minimal parameters that need to be adapted depending on
103 the environment and system the user is using, which are all clearly explained in the Github
104 repository. Finally, the Github repository also includes an example of how to execute the
105 pipeline using a SLURM workload manager (*run_nextflow.sh*).

Research impact statement

107 The pipeline was initially developed for internal use purposes, as repeated analyses across
108 dozens of GWAS summary statistics had become a systematic task. Since then, new members
109 of our research group (junior and senior level) have tested and successfully utilized the pipeline,
110 without having previously used any of the tools contained in the pipeline, in short time. The
111 pipeline has been available as a public Github repository since July 2025 and it has already been
112 starred by external Github users, attesting to its utility for the field. We foresee that our pipeline
113 will become the go-to approach for computational genetics researchers who may want to include
114 genetic correlations as primary or secondary analysis to answer their research questions using
115 GWAS summary statistics. Additionally, our pipeline facilitates the implementation of these
116 tools by a diverse set of researchers, including clinician-scientists, who may not have sufficient
117 expertise in bioinformatics to run genetic correlation analyses using the original tools as is.

AI usage disclosure

119 Generative AI (Claude Code) was used to create the R environment Apptainer image, followed by
120 tests across different test users and system environments to ensure its functionality. Generative
121 AI (Claude Code) was also used to create a draft of the documentation available on Github,
122 which was then manually updated and improved by the authors.

- 168 Wu, X., Shadrin, A., Blokland, G. A. M., Guloksuz, S., Andreassen, O. A., Linden, D. E. J., &
169 Van Der Meer, D. (2026). Shared genetic architecture between alzheimer's disease and
170 brain morphology. *Alzheimer's Research & Therapy*. [https://doi.org/10.1186/s13195-026-](https://doi.org/10.1186/s13195-026-02105-5)
171 [02105-5](https://doi.org/10.1186/s13195-026-02105-5)
- 172 Zhang, W., Zhang, Q., Luo, Y., Ma, L., Wang, X., Zheng, Q., Zhang, X., Wu, S., Li, Z.,
173 & Bi, Y. (2025). Genomic correlation, shared loci, and causal link between obesity and
174 diabetic microvascular complications: A genome-wide pleiotropic analysis. *Biomolecules*
175 *and Biomedicine*, 25(10), 2197–2208. <https://doi.org/10.17305/bb.2025.11897>

DRAFT